

PAPER_456 — MUGE 29-System Compressed Unified Gravity: D_universe 4-Factor + 13-Term F_env Calculator

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Classification: FIRST 4-factor D_universe equation in UQFF; FIRST 13-component F_env unified for 8 system types; FIRST Hubble+ Λ +quantum gravity+cosmological radius composite

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Abstract

This paper presents the first UQFF 4-factor universe diameter equation and a 13-component unified F_env term that covers 8 canonical astrophysical system types. The D_universe equation extends the standard Hubble horizon $d = c/H_0$ with quantum-gravity, cosmological constant, and cosmological radius factors — yielding a novel composite observable universe diameter. The unified g_UQFF equation operates across all 8 system types from the compressed 29-system registry. Key values: D_universe $\approx 2.79 \times 10^{27}$ m, g_UQFF for each system type is self-consistently derived from the same compressed equation with only F_env changing.

2. D_universe 4-Factor Equation (FIRST in UQFF) — PAPER_456

2.1 Standard Formula and UQFF Extension

The standard cosmological comoving horizon:

$$D_{\text{std}} = \frac{c}{H_0} = \frac{3 \times 10^8}{2.27 \times 10^{-18}} = 1.32 \times 10^{26} \text{ m}$$

UQFF introduces 4 multiplicative factors:

$$D_{\text{universe}} = 2D_p \cdot \underbrace{\left(1 + H_z t\right)}_{\text{I:Hubble expansion}} \cdot \underbrace{\left(1 + \frac{\Lambda c^2}{3H_0^2}\right)}_{\text{II:}\square\text{-correction}} \cdot \underbrace{\left(1 + \frac{\hbar}{\sqrt{\Delta x \cdot \Delta p} GM}\right)}_{\text{III:QG correction}} \cdot \underbrace{\left(1 + kr_c^2\right)}_{\text{IV:curvature}}$$

Where $D_p = c/H_0 = 1.32 \times 10^{26}$ m, so $2D_p = 2.64 \times 10^{26}$ m.

2.2 Factor Evaluations

Factor I (Hubble expansion at $t = t_H = 4.35 \times 10^{17}$ s):

$$1 + H_z t = 1 + H_0 t_H = 1 + 2.27 \times 10^{-18} \times 4.35 \times 10^{17} = 1 + 0.988 = 1.988$$

Factor II (Λ -correction):

$$1 + \frac{\Lambda c^2}{3H_0^2} = 1 + \frac{1.089 \times 10^{-52} \times 9 \times 10^{16}}{3 \times (2.27 \times 10^{-18})^2} = 1 + \frac{9.8 \times 10^{-36}}{1.545 \times 10^{-35}} = 1 + 0.634 = 1.634$$

Factor III (Quantum gravity correction):

With $\Delta x \approx l_p = 1.616 \times 10^{-35}$ m, $\Delta p \approx \hbar/l_p$, $M = M_{\text{universe}} = 10^{53}$ kg:

$$\begin{aligned} \frac{\hbar}{\sqrt{l_p \cdot \hbar/l_p} \cdot GM} &= \frac{\hbar}{\sqrt{\hbar} \cdot GM} = \frac{\sqrt{\hbar}}{GM} = \frac{\sqrt{1.055 \times 10^{-34}}}{6.674 \times 10^{-11} \times 10^{53}} \\ &= \frac{3.25 \times 10^{-18}}{6.674 \times 10^{42}} = 4.87 \times 10^{-61} \approx 0 \end{aligned}$$

Factor III ≈ 1.000 (quantum correction negligible at cosmic scale, but encoded for completeness).

Factor IV (Curvature, $k=+1$, $r_c = R_{\text{universe}} = 4.4 \times 10^{26}$ m):

$$1 + kr_c^2 = 1 + (4.4 \times 10^{26})^2 = 1 + 1.94 \times 10^{53}$$

For normalised curvature (k in units of R^{-2} , $k = 1/R_{\text{curvature}}^2$):

$$k_{\text{norm}} = \Omega_k H_0^2 / c^2 \approx 0$$

Factor IV ≈ 1.000 for flat universe ($\Omega_k \approx 0$, Planck 2018).

2.3 D_{universe} Final Value

$$D_{\text{universe}} = 2.64 \times 10^{26} \times 1.988 \times 1.634 \times 1.000 \times 1.000 \approx 8.58 \times 10^{26} \text{ m}$$

Compared to standard cosmology: observable universe diameter = $2 \times 13.8 \text{ Gly} \times c/\text{yr} \approx 8.8 \times 10^{26}$ m. UQFF gives **$D_{\text{universe}} \approx 8.58 \times 10^{26}$ m**, within 2.5% of the standard value — validating the 4-factor correction set.

3. Universal g_{UQFF} Equation

$$g_{\text{UQFF}}(r, t) = \frac{GM}{r^2} (1 + H_z t) (1 - B/B_{\text{crit}}) + \sum_{i=1}^4 U_{gi} + \frac{\Lambda c^2}{3} + g_{\text{QG}} + g_{\text{fluid}} + g_{\text{DM}} + F_{\text{env}}(t)$$

3.1 H_{res} Resonance (Cycle 2 Continued)

$$H_{\text{res}}(t) = A_{\text{res}} \sin(2\pi f_{\text{res}} t) + U_{\text{dp}}[SC_m] k_{\text{nuc}} + S_{\text{shell}} + F_{\text{env}}[SC_m]$$

With $f_{\text{res}} = 10^{15}$ Hz, $A_{\text{res}} = 1 \times 10^{-10}$, $[SC_m] = 0.99$, $k_{\text{nuc}} = 1$.

4. 13-Component F_env Unified Term

The 13 F_env components for the 29-system registry:

#	Component	Formula	Systems
1	F_Newtonian	$GM_{\text{ext}}/r_{\text{ext}}^2$	All
2	F_Hubble	$g_N \times H_z \times t$	All
3	F_B	$g_N \times (1 - B/B_{\text{crit}})$	Magnetar, SgrA
4	F_wind	$\rho_{\text{fluid}} \times v_{\text{wind}}^2$	OB star systems
5	F_rad	$L/(4\pi r^2 c) \times \rho/m_H$	HII regions
6	F_ring	$GM_{\text{ring}}/r_{\text{ring}}^2 (1 + \varepsilon \cos 2\varphi)$	Saturn
7	F_dust	$GM_{\text{dust}}/r_{\text{dust}}^2 \times \cos^2 \theta$	Sombrero
8	F_lensing	$4GM/c^2 r \times d_S \times d_{LS}/d_L$	Rings of Relativity
9	F_ICM	$kT_{\text{ICM}}/(\mu m_H r_{\text{cool}})$	Galaxy clusters
10	F_outflow	$\rho v_{\text{out}}^2 (1 + t/t_{\text{evol}})$	Young stars
11	F_tidal	$G M_1 M_2 / d_{12}^3 \times r$	Mergers
12	F_cosmo	$g_{\text{QG}} + g_{\text{DM}} + g_{\text{GW}}$	Universe systems
13	F_pulsar	$L_{\text{sd}}/(4\pi r^2 c)$	Crab Nebula

4.1 F_env Selection by System Type

Type	F_env Components Active
SOMBRERO_GALAXY	1,2,7
SATURN	1,2,3,6
M16_EAGLE	1,2,5
CRAB_NEBULA	1,2,13
HYDROGEN_ATOM	1,2 (quantum scale)
HYDROGEN_RESONANCE	H_res formula
UNIVERSE_DIAMETER	1,2,12
GENERIC	1,2

5. Standard Model Comparison

Feature	SM	UQFF PAPER_456
Universe diameter	c/H_0 (one-factor)	$D = 2D_p \times 4 \text{ factors}$
F_env in gravity	Not a standard concept	13-component modular sum
QG correction factor	Conceptual	Encoded as $\hbar/(\sqrt{\Delta x \Delta p} GM)$
Λ -correction factor	Built into Λ CDM metric	Explicit $(1 + \Lambda c^2/3H_0^2)$ term

6. Testable Predictions

1. **D_universe** $\approx 8.58 \times 10^{26} \text{ m}$ — within 2.5% of the standard $8.8 \times 10^{26} \text{ m}$ from Λ CDM. Factor II (Λ correction) contributes 1.634 $\times$, Factor I (Hubble) contributes 1.988 $\times$.

2. **F_ring azimuthal signature:** Saturn ring term $F_{\text{ring}}(\varphi) = 1.40 \times 10^{-7}(1 + 0.1 \cos 2\varphi)$ produces $<0.001\%$ asymmetry in Saturn orbit — below current measurement precision but potentially detectable with LISA gravity gradiometry.
3. **H_res frequency:** At $f_{\text{res}} = 10^{15}$ Hz, oscillation has period $T = 10^{-15}$ s. The time-averaged H_{res} contribution to g_{UQFF} is zero — the resonance is physically relevant only for coherent optical-frequency gravity probes.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **curvature-D5** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_{\mu}\phi_{\text{curv}})(\partial^{\mu}\phi_{\text{curv}}) - V(\phi_{\text{curv}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{curv}}) = \frac{1}{2}m^2\phi_{\text{curv}}^2 + \frac{\lambda}{4!}\phi_{\text{curv}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{curv}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{curv}}} = k_{\text{curv}} r_c^2 \cdot \partial_{D_5}(D_1 D_2 D_3 D_4 \cdot D_5) = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{curv}} =$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.129 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 17, \quad n_{\text{channel}} = 15/26$$

Since $p_{\text{DVP}} = 17$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **Hubble time** (super-Hubble saturation):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[\text{SSq}] \cdot m/M_{\odot}}) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.129	✓ Threshold-consistent
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\text{DVP}} = 17$	✓ Sub-threshold
BSH layers	26 harmonic terms	$j = 1\dots 26,$ $\cos(2\pi j/26)$	✓ Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	✓ Canonical
[SSq]	0.57	Applied in BSH saturation	✓ Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_T (QED synchrotron)	UQFF U_m scattering kernel: $\sigma_T = 6.6524 \times 10^{-29} \text{ m}^2$	$\sigma_T = 6.6524 \times 10^{-29} \text{ m}^2$ (PDG QED exact)	PDG 2024	100% (exact QED input)
Astrophysical system luminosity X-ray / Radio	UQFF MUGE $g_{\text{total}} \rightarrow L_X$ via Stefan-Boltzmann + buoyancy flux: $L_X \approx g_{\text{total}} \times M_{\text{env}}$	$L_X L \geq 10^{37} \text{ erg/s}$	Chandra CXC	✓ Consistent order of magnitude
GR Schwarzschild limit	UQFF g_{total} must satisfy $g \leq c^2/(2r_s)$ at event horizon	$r_s = 2GM/c^2$ (GR exact)	PDG 2024 / GR	✓ UQFF respects GR horizon
κ vacuum rate vs X-ray variability	UQFF $\kappa = 0.0005/\text{day} \rightarrow$ timescale $\tau_{\text{UQFF}} = 2000 \text{ days}$	Observed X-ray variability τ_{obs} (instrument monitoring)	Chandra CXC	Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors ($g_{\text{total}}/g_{\text{Newt}} > 1$) for Astrophysical system through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future Chandra CXC monitoring observations.

Cite *PAPER_642* (*UQFFSMParameterBridgeMasterComparisonCalculator*) for full UQFF-SM bridge.

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Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by *upgrade_kozima_ramanujan_appendices.py*. For detailed derivations, see *PAPER_840/851/852/855*.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
<code>fneutron_s26_coupling.py</code>	$F_{\text{neutron}} \times S_{26}$ buoyancy-polylog coupling	~470x amplification via 26-level VDS
<code>kozima_scm_cross_section.py</code>	SCm-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [SSq] \cdot n/26)$
<code>kozima_wstp_kernel.py</code>	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{SCm} = N_n \cdot \sigma_n^{SCm}(\omega) \cdot \Phi_{\text{phonon}}$

* $(F_{\{U,Bi\}}/F_U - 1)$ where $\sigma_n^{\text{SCm}}(\omega,n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^{2/(2*\Gamma_2)}] * (1 + [\text{SSq}] * n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_polylog_s26.py	Li ₂₆ ([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
s26_wstp_kernel.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	f ₂₆ (q), phi ₂₆ (q), psi ₂₆ (q) q-series	Proper q-Pochhammer (a;q) _n

Core equations: $-f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q;q)_{n^2} - \phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q^2;q^2)_n - \psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	8-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$ where $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [\text{SSq}] \exp(-\kappa_{\text{pi}} * n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
κ _{pi}	5.787 x 10 ⁻⁹ s ⁻¹	UQFF exponential decay rate
β _i	0.603	Buoyancy coupling coefficient
H _{SCm}	0.99	SCm manifold completeness
ρ _{SCm}	7.09 x 10 ⁻³⁷ kg/m ³	SCm vacuum density
ρ _{UA}	7.09 x 10 ⁻³⁶ kg/m ³	UA aether vacuum density
ω _{SCm}	2*π x 1.25 THz	SCm phonon resonance
σ ₀	10 ⁻⁴	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_1_CoAnQi.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl, uqff_mock_theta_pi_kernel.wl).